AI-Powered Personal Finance Management and Budget Recommendation System Using Deep Learning

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

MLA0408 - DEEP LEARNING

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**AI-Powered Personal Finance Management and Budget Recommendation System Using Deep Learning"** has been carried out by **Nalipi Reddy Rohith (192425115)**,**G.J.D.Sai Sharan (192425152)**,**K Giridhar (192425160)**,**S Syed Abuzar(192425384)**under the supervision of **Dr Sudhagar D** and is submitted in partial fulfilment of the requirements for the current semester of the **Computer Science and Engineering** (**B.Tech)** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR	COURSE FACULTY
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DECLARATION

We, **Nalipi Reddy Rohith (192425115)**,**G.J.D.Sai Sharan (192425152)**,**K Giridhar (192425160)**,**S Syed Abuzar(192425384)**of the **[B.Tech AI&ML]**, SIMATSEngineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "**AI-Powered Personal Finance Management and Budget Recommendation System Using Deep Learning**" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Nalipi Reddy Rohith(192425115)
G.J.D.Sai Sharan (192425152)
K Giridhar (192425160)
S Syed Abuzar(192425384)

Signature of the Students with Names

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Individual Contribution Statement

Student	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Contribution
Nalipi Reddy Rohith(192425115)	User and financial data management	Financial data collection and preprocessing	Data validation and testing	Chapters 1–2	**25%**
G.J.D.Sai Sharan (192425152)	Smart expense classification and analytics	Expense classification and spending analysis	Classification accuracy analysis	Chapter 3	**25%**
K Giridhar (192425160)	Budget prediction and recommendation	Deep learning-based budget prediction	Prediction and recommendation evaluation	Chapter 4	**25%**
S Syed Abuzar(192425384)	Financial health monitoring and reporting	Financial dashboard and report generation	Performance evaluation and result analysis	Chapter 5	**25%**
Total					**100%**

ABSTRACT

Personal finance management is an important challenge for individuals who need to effectively control their income, expenses, savings, and financial goals. Traditional methods of financial management mainly depend on manual tracking and fixed budgeting rules, which may not adequately consider individual spending behavior. The proposed **AI-Powered Personal Finance Management and Budget Recommendation System Using Deep Learning** addresses this problem by providing an intelligent system for financial data management, expense analysis, budget prediction, personalized recommendations, and financial health monitoring.

The proposed system consists of four major modules. The first module manages user and financial data, including income, expenses, savings, investments, loans, and financial goals. The second module performs smart expense classification and analytics by categorizing transactions and identifying spending patterns. The third module uses deep learning techniques to analyze historical financial data, predict future expenses, and generate personalized budget recommendations. The fourth module monitors financial health by analyzing income, expenditure, savings, and budget adherence and generates financial reports and insights.

The system aims to help users understand their financial behavior, identify unnecessary spending, maintain appropriate budgets, improve savings, and make informed financial decisions. By combining financial data processing with deep learning-based prediction and recommendation, the proposed system provides a personalized approach to personal finance management.

Keywords: Personal Finance, Deep Learning, Budget Prediction, Expense Classification, Financial Analytics, Recommendation System

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES OF TABLES

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
DL	Deep Learning	—
ML	Machine Learning	—
PM	Personal Finance Management	—
ECR	Expense-to-Income Ratio	%
SR	Savings Rate	%
CNN	Convolutional Neural Network	—
RNN	Recurrent Neural Network	—
LSTM	Long Short-Term Memory	—
RMSE	Root Mean Square Error	—
MAE	Mean Absolute Error	—
MSE	Mean Squared Error	—
₹	Indian Rupee	INR
%	Percentage	%
INR	Indian Rupee	—
API	Application Programming Interface	—
UI	User Interface	—
DB	Database	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

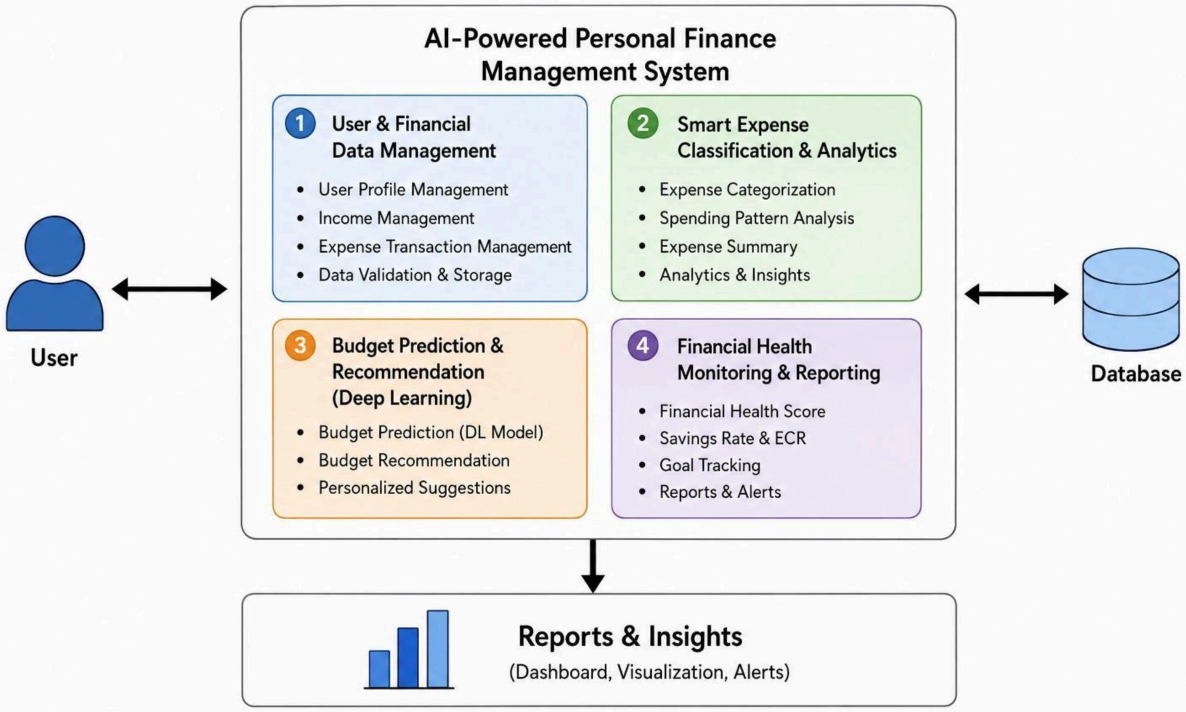
Personal finance management involves planning and controlling an individual's income, expenses, savings, investments, loans, and financial goals. With the increasing use of digital payments and online transactions, individuals generate a large amount of financial data. However, managing and analyzing this information manually can be difficult and time-consuming.

Many users are unable to accurately understand their spending patterns or determine how much they should allocate to different expense categories. Conventional budgeting approaches generally use fixed limits and do not sufficiently adapt to changes in individual financial behavior.

Artificial Intelligence and Deep Learning provide opportunities to analyze large amounts of financial data and identify complex patterns. By learning from historical income and expense information, an intelligent system can predict future spending and generate personalized budget recommendations.

The proposed project combines financial data management, expense classification, analytics, deep learning-based prediction, budget recommendation, and financial health monitoring into one integrated system.

Figure 1.1 Overall System Architecture



1.2 Need for the Project

The project is needed because:

- Manual financial tracking can be time-consuming.
- Users may have difficulty categorizing expenses.
- Unnecessary spending may not be easily identified.
- Fixed budgets may not match individual spending behavior.
- Future expenses are difficult to estimate manually.
- Users require personalized budget recommendations.
- Financial health needs continuous monitoring.
- Historical financial data can be utilized for intelligent predictions.

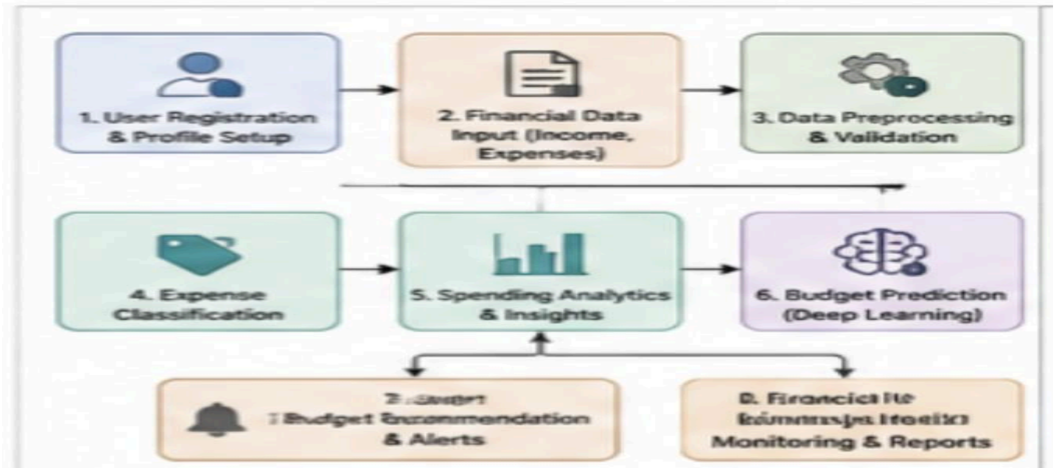


Figure 1.2: Proposed System Workflow

1.3 Problem Statement

To design and develop an AI-powered personal finance management system that collects and analyzes financial data, automatically classifies expenses, predicts future spending using deep learning, recommends personalized budgets, and monitors the user's financial health through analytical reports.

1.4 Project Aim

To develop an intelligent personal finance management and budget recommendation system using deep learning for analyzing financial behavior, predicting expenses, recommending suitable budgets, and improving users' financial planning.

1.5 Project Objectives

1. To design a system for collecting and managing user financial information.
2. To develop a mechanism for recording and categorizing income and expenses.
3. To analyze user spending patterns and identify major expenditure categories.
4. To implement smart expense classification and financial analytics.
5. To develop a deep learning model for predicting future expenses.
6. To generate personalized budget recommendations based on financial behavior.
7. To monitor financial health using income, expenditure, and savings indicators.
8. To generate financial reports and visualizations.
9. To test and evaluate the performance of the proposed system.

1.6 Scope

The project includes:

- User financial data management
- Income and expense tracking
- Expense classification
- Spending analytics
- Historical financial data preprocessing
- Deep learning-based expense prediction
- Budget recommendation
- Savings-related recommendations
- Financial health monitoring
- Financial reports and visualizations

The system does not include direct execution of banking transactions or guaranteed investment returns.

1.7 Expected

Engineering

Outcome

The expected outcome is a software-based intelligent personal finance management prototype capable of collecting financial information, analyzing spending behavior, predicting future expenses, recommending budgets, and generating financial health reports.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review focuses on existing personal finance management applications, expense classification approaches, financial prediction techniques, machine learning and deep learning methods, and recommendation systems.

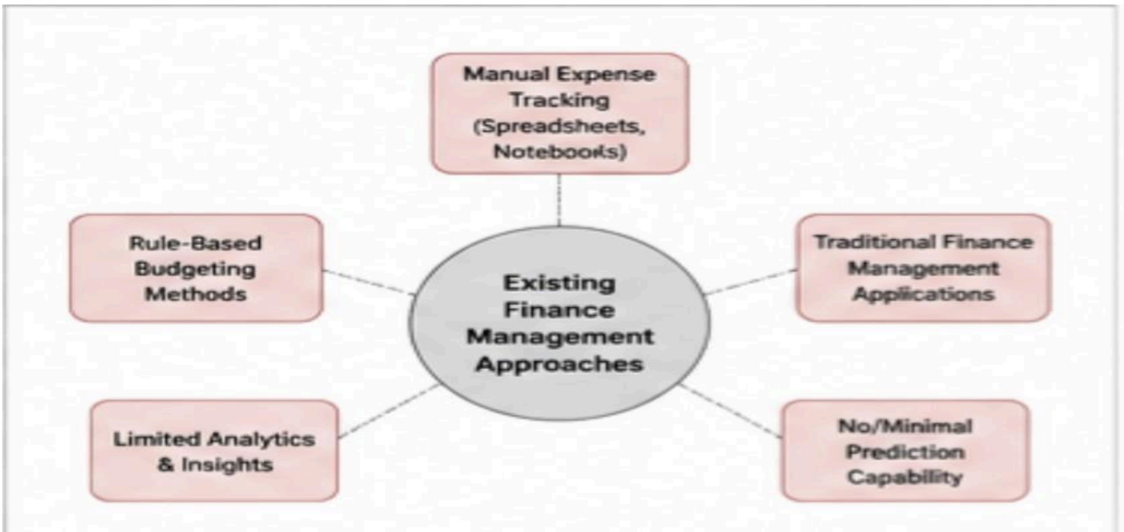


Figure 2.1: Existing System Approaches

2.2 Review of Existing Work

Manual Financial Management

Traditional financial management uses notebooks, spreadsheets, or manually maintained records. These methods are simple but require continuous user effort and provide limited automated analysis.

Rule-Based Budgeting

Rule-based systems assign budgets using predefined financial rules. Although easy to implement, these systems may not sufficiently adapt to individual spending patterns.

Personal Finance Applications

Existing applications provide features such as expense recording, transaction categorization, budget tracking, and financial reports. However, the level of personalized prediction and recommendation varies between systems.

Machine Learning-Based Financial Analysis

Machine learning can analyze historical financial data and identify spending patterns. It can improve automated classification and prediction compared with purely rule-based systems.

Deep Learning-Based Prediction

Deep learning can learn complex patterns from historical financial information and can be applied to expense forecasting and personalized budget recommendation.

Feature	Manual Methods	Traditional Apps	Proposed System
Expense Tracking	✓	✓	✓
Automated Classification	×	Partial	✓
Spending Analytics	×	Partial	✓
Budget Prediction (Deep Learning)	×	×	✓
Personalized Recommendations	×	Partial	✓
Financial Health Monitoring	×	Partial	✓
Real-time Insights	×	Partial	✓

Figure 2.2: Comparative Analysis

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance
Manual Tracking	Simple	Time-consuming	Basic financial recording
Rule-Based Budgeting	Easy to implement	Less personalized	Budget baseline
Traditional Finance Apps	Expense tracking and reports	Limited intelligent prediction	Financial management
Machine Learning	Pattern recognition	Requires quality data	Financial analysis
Proposed System	Integrated AI-based prediction and recommendation	Requires sufficient historical data	Complete solution

2.4 Research / Engineering Gap

Existing approaches often focus primarily on expense recording and historical reporting. A more integrated approach is required to combine financial data management, automatic expense classification, spending analytics, deep learning-based prediction, personalized budget recommendation, and financial health monitoring.

2.5 Proposed Contribution

The proposed system integrates four major modules into a single personal finance management framework:

Financial Data → Expense Classification → Deep Learning Prediction → Budget Recommendation → Financial Health Monitoring

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Functional Requirements

Requirement	Description
User Management	Manage user financial profile
Income Management	Record and track income
Expense Management	Record and track expenses
Expense Classification	Categorize transactions
Financial Analytics	Analyze spending patterns
Expense Prediction	Predict future expenses
Budget Recommendation	Recommend category-wise budgets
Financial Health Monitoring	Monitor financial indicators
Reporting	Generate financial reports

Non-Functional Requirements

Requirement	Target
Usability	Simple and understandable interface
Reliability	Consistent financial calculations
Security	Protect financial information
Performance	Provide timely analysis
Scalability	Support increasing transaction records

3.2 Constraints & Applicable Standards

Standard / Requirement	Application

Data privacy principles	Protection of personal financial information
Secure software practices	Authentication and controlled access
Academic integrity requirements	Proper acknowledgement of datasets, software, and resources

3.3 Design Specifications

Parameter	**Target**
Financial data management	Structured
Expense classification	Automated
Financial prediction	Deep learning based
Budget recommendation	Personalized
Reporting	Category-wise and monthly
Security	Authorized access
Data processing	Clean and model-ready

3.4 Alternative Solutions & Evaluation

Alternative 1 — Manual System

Uses spreadsheets or manual records.

Alternative 2 — Rule-Based System

Uses predefined budgeting and classification rules.

Alternative 3 — AI-Based System

Uses financial data analytics and deep learning to predict spending and recommend budgets.

Criterion	**Weight**	**Manual**	**Rule-Based**	**AI-Based**
Performance	25%	Medium	Medium	High
Personalization	25%	Low	Medium	High
Automation	20%	Low	Medium	High
Scalability	15%	Low	Medium	High
Prediction	15%	Low	Low	High

Selected: AI-Based System.

3.5 Selected Design & Detailed Design

Proposed Architecture

User

↓

User & Financial Data Management

↓

Data Preprocessing

↓

Smart Expense Classification & Analytics

↓

Deep Learning-Based Budget Prediction

↓

Personalized Budget Recommendation

↓

Financial Health Monitoring

↓

Reports & Dashboard

The selected design integrates all four modules so that the output of one component becomes the input for the next stage.

3.6 Trade-Off Analysis

Design Decision	**Alternative**	**Benefit**	**Disadvantage**	**Final Decision**
Automated classification	Manual classification	Saves time	Requires trained model/rules	Automated
Deep learning prediction	Fixed rules	More adaptive	Requires data	Deep learning
Personalized budget	Fixed budget	User-specific	Requires user history	Personalized
Digital reporting	Manual reports	Faster analysis	Requires visualization system	Digital

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Impact
Sustainability	Digital financial records	Reduces paper-based records
Ethics	Responsible handling of financial information	Protect user privacy
Safety	Software-based system	No physical safety hazard
Security	Financial information protection	Authentication and controlled access

Risk Register

Risk	Likelihood	Severity	Mitigation
Incorrect user data	Medium	Medium	Input validation
Missing transaction data	Medium	Medium	Data preprocessing
Insufficient training data	Medium	High	Increase dataset size
Incorrect prediction	Medium	High	Model evaluation
Unauthorized access	Low	High	Authentication and access control

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The proposed system is developed using a modular architecture. Financial information is first collected and preprocessed before being analyzed by the expense classification module. The resulting financial patterns are provided to the deep learning prediction component, which estimates future expenditure and supports personalized budget recommendations. The final module monitors financial health and presents analytical reports to the user.

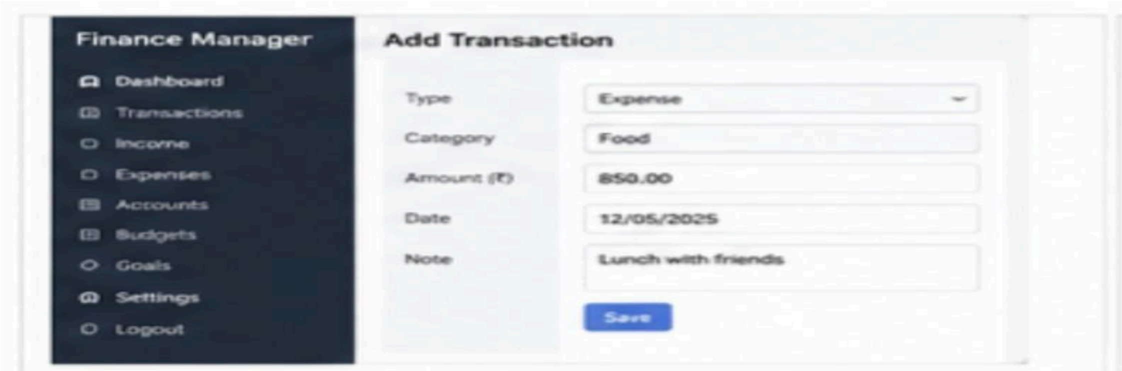


Figure 4.1: User & Financial Data Management Interface

4.2 Software Implementation & Modern Tools Used

Tool / Software	Purpose
Python	System development
Pandas	Data preprocessing
NumPy	Numerical operations
Database	Financial data storage
Deep Learning Framework	Prediction model
Matplotlib / Visualization Tools	Financial charts
Web/Application Framework	User interface

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
User management	Enter user details	Data stored correctly

Expense recording	Add transactions	Transactions recorded
Classification	Test transactions	Correct category assigned
Analytics	Analyze transaction data	Spending summary generated
Prediction	Test historical data	Future expense predicted
Recommendation	Provide financial profile	Budget generated
Reporting	Generate report	Financial report displayed

4.4 Results

The proposed system is expected to provide:

- Structured user financial records.
- Automated expense categorization.
- Category-wise spending analysis.
- Identification of major spending areas.
- Future expense predictions.
- Personalized budget recommendations.
- Savings-related insights.
- Financial health indicators.
- Monthly and category-wise financial reports.

Important: For the final report, actual accuracy, prediction error, graphs, screenshots, and test values should be inserted here based on your implemented system. The sample specifically asks for measurements, graphs, and key data in the Results section.

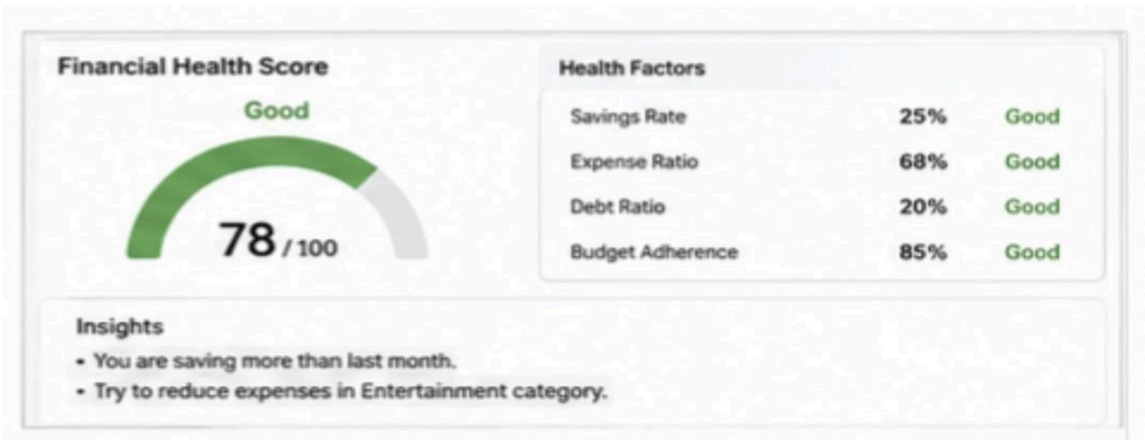


Figure 4.4: Financial Health Monitoring Dashboard

4.5 Data Analysis & Engineering Judgment

The performance of the system depends strongly on the quality and quantity of financial data. Proper preprocessing and expense classification are necessary before applying the deep learning model. Prediction performance should be evaluated using appropriate error or accuracy metrics. The recommendations should be evaluated based on whether the generated budget is consistent with the user's income, historical expenditure, and financial goals.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Manage financial data	Financial data module	Fully/Partially
Classify expenses	Expense classification	Fully/Partially
Analyze spending	Analytics module	Fully/Partially
Predict expenses	Deep learning model	Fully/Partially
Recommend budgets	Recommendation module	Fully/Partially
Monitor financial health	Monitoring module	Fully/Partially
Generate reports	Reporting dashboard	Fully/Partially

4.7 Strengths & Limitations

Strengths

- AI-based financial analysis.
- Personalized budget recommendations.
- Automated expense classification.
- Future expense prediction.
- Financial health monitoring.
- Integrated financial reporting.
- Supports better financial decision-making.

Limitations

- Prediction performance depends on historical data quality.
- Insufficient transaction history may reduce prediction reliability.
- Incorrect user input can affect recommendations.
- The system provides decision-support recommendations and does not guarantee financial outcomes.

4.8 Project Planning, Roles & Budget

Project Modules

Module	Responsibility
Module 1	User & Financial Data Management
Module 2	Smart Expense Classification & Analytics
Module 3	Deep Learning-Based Budget Prediction & Recommendation
Module 4	Financial Health Monitoring & Reporting

Project Development Stages

Requirement Analysis → Data Collection → Data Preprocessing → Module Development → Deep Learning Model → Integration → Testing → Evaluation → Documentation

-

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The **AI-Powered Personal Finance Management and Budget Recommendation System Using Deep Learning**provides an integrated approach to personal financial management. The system combines user and financial data management, smart expense classification, financial analytics, deep learning-based budget prediction, personalized recommendations, and financial health monitoring.

The system aims to overcome the limitations of traditional manual and fixed-rule budgeting methods by using historical financial behavior to generate more personalized insights. By analyzing income and expenditure patterns, the system can support users in controlling unnecessary spending, maintaining appropriate budgets, and improving their financial planning.

Overall, the proposed system demonstrates how Artificial Intelligence and Deep Learning can be applied to personal finance management to provide automated, data-driven, and personalized financial assistance.

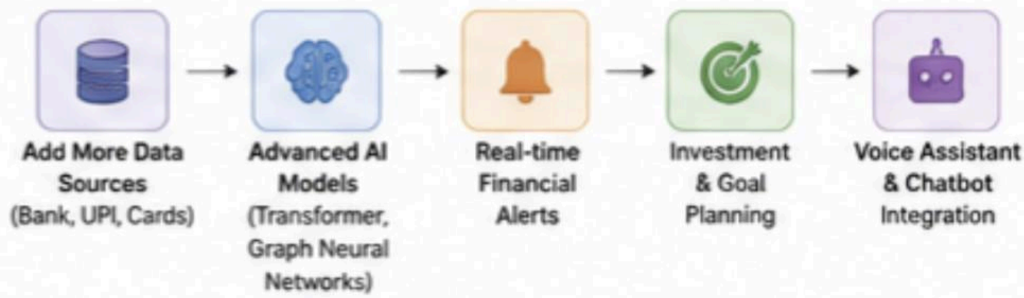


Figure 5.1: Future Enhancement Workflow

5.2 Future Work

Future enhancements can include:

- Real-time banking transaction integration.
- Automatic transaction collection.
- Advanced deep learning architectures.
- Improved expense classification.
- Real-time financial alerts.
- Investment planning features.
- Fraud and abnormal spending detection.
- Mobile application development.
- Multilingual support.
- Improved personalized financial goals.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Personal financial data management
- Expense analytics
- Deep learning
- Financial prediction
- Recommendation systems
- Data visualization
- Software system integration

Engineering & Professional Skills Developed

- Problem analysis
- Data preprocessing
- Machine learning/deep learning implementation
- Software development
- Testing and validation
- Technical documentation
- Team collaboration

Individual Reflection

During the development of this project, I gained practical knowledge of applying Artificial Intelligence and Deep Learning to a real-world financial management problem. I learned how financial data can be collected, processed, analyzed, and transformed into useful predictions and recommendations. Working on the project also improved my understanding of software design, model evaluation, testing, and technical documentation. The project helped me understand the importance of data quality, system security, and user-specific requirements when developing an intelligent application.

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APPENDICES

Appendix A – Detailed Calculations

This appendix contains the important calculations used in the proposed **AI-Powered Personal Finance Management and Budget Recommendation System Using Deep Learning**.

The calculations include:

- Total monthly income calculation.
- Total monthly expense calculation.
- Category-wise expense calculation.
- Monthly savings calculation.
- Savings rate calculation.
- Expense-to-income ratio.
- Budget allocation calculations.
- Prediction performance calculations such as MAE, MSE, RMSE, or other applicable evaluation metrics.

Appendix B – Engineering Drawings / Schematics

This appendix contains the system architecture and workflow diagrams of the proposed system. The major components represented in the system architecture are:

User Input → User & Financial Data Management → Data Preprocessing → Smart Expense Classification & Analytics → Deep Learning-Based Budget Prediction & Recommendation → Financial Health Monitoring & Reporting

The diagrams illustrate the interaction and data flow between the different modules of the proposed system.

Appendix C – Source Code / Code Repository Information

This appendix contains essential source-code excerpts used for implementing the proposed system.

The important code components include:

- User and financial data management.
- Data preprocessing.
- Expense classification.
- Financial analytics.
- Deep learning model implementation.
- Budget prediction.
- Budget recommendation.
- Financial health calculation.
- Report and dashboard generation.

The complete source code may be maintained separately in the approved project repository. Only essential code excerpts and implementation details are included in the report.

Appendix D – Datasheets

This appendix contains the technical information related to the software tools, frameworks, libraries, datasets, and other resources used in the project.

The resources include:

- Python development environment.
- Pandas and NumPy libraries.
- Deep learning framework.
- Data visualization libraries.
- Database technologies.
- Financial dataset used for training and testing.
- Other open-source resources used during development.

Appendix E – Experimental / Raw Data

This appendix contains the raw and processed financial data used for developing and evaluating the system.

The data may include:

- User financial profiles.
- Monthly income records.
- Expense transactions.
- Expense categories.
- Savings information.
- Financial goals.
- Historical spending records.
- Training and testing datasets.
- Model prediction results.
- Budget recommendation outputs.

Appendix F – Ethics / Institutional Approval

The project involves handling financial information, which requires responsible data management and protection of user privacy.

The system is designed to:

- Avoid unnecessary collection of personal information.
- Protect financial information from unauthorized access.
- Use financial data only for the intended analysis.
- Maintain responsible handling of datasets.
- Clearly acknowledge external datasets, software, and AI-assisted resources used in the

project.

Institutional approval documentation will be included if applicable.

Appendix G – Project Meeting / Progress Records

This appendix contains records of project discussions, development progress, review meetings, and milestones.

The records may include:

- Project topic selection.
- Requirement analysis.
- Module planning.
- Dataset selection.
- System design discussions.
- Development progress.
- Testing and evaluation discussions.
- Final project review.

Appendix H – User / Industry Feedback

This appendix contains feedback obtained from users or relevant stakeholders regarding the proposed personal finance management system.

The feedback may cover:

- Ease of entering financial information.
- Ease of understanding expense categories.
- Usefulness of spending analytics.
- Relevance of budget recommendations.
- Understanding of financial health reports.
- Suggestions for improving the

system. This section will be included where

applicable.

Appendix I – Publications / Patents / Competitions / Product Outcomes

This appendix contains details of any publications, patents, competitions, demonstrations, or product outcomes resulting from the project.

Where applicable, the following information may be included:

- Research paper publication.
- Conference presentation.
- Project competition participation.

- Prototype demonstration.

- Product or application outcome.
- Awards or recognitions.

If none are applicable, this appendix may be marked as **Not Applicable**.

Appendix J – Individual Contribution Evidence

This appendix provides evidence of the individual contribution of **Nalipi Reddy Rohith (192425115)** and other team members, where applicable.

The evidence may include:

- Module development records.
- Source-code contributions.
- System design contributions.
- Testing and analysis records.
- Screenshots of implementation work.
- Report-writing contributions.
- Project meeting records.
- Module-wise responsibilities.